

Network Security Assessment – Scan Methodology

1. Introduction

This document describes the methodology used for conducting the Network Security Assessment. The assessment focuses on identifying network services, evaluating potential security weaknesses, and documenting risks through a structured audit approach.

The methodology follows a controlled security assessment process designed to evaluate network visibility and security posture.

2. Assessment Objectives

The objectives of the network scan are:

- Identify active devices within the assessment environment.
 - Detect open ports and running services.
 - Collect information about network configurations.
 - Identify potential security weaknesses.
 - Support risk assessment and remediation planning.
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3. Assessment Approach

The assessment was conducted using the following phases:

Phase 1: Planning and Preparation

Activities performed:

- Defined assessment scope.
 - Identified target environment.
 - Selected appropriate scanning techniques.
 - Prepared documentation templates.
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Phase 2: Network Discovery

Network discovery activities include:

- Identifying active hosts.
- Reviewing available network devices.
- Understanding network exposure.

Purpose:

To determine which systems are accessible within the assessment environment.

Phase 3: Port and Service Scanning

The assessment identifies:

- Open ports
- Running services
- Service versions
- Potentially unnecessary exposed services

Purpose:

To identify possible entry points that may increase security risk.

Phase 4: Vulnerability Analysis

The identified services are reviewed against security references to determine:

- Known vulnerabilities
- Misconfigurations
- Security weaknesses

Sources used:

- Common Vulnerabilities and Exposures (CVE)
 - Security best practices
 - Vendor recommendations
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4. Tool Used

Nmap (Network Mapper)

Nmap is used for network discovery and security auditing.

Functions performed:

Function	Purpose
Host Discovery	Identify active devices
Port Scanning	Identify open ports
Service Detection	Identify running services
Version Detection	Collect service information

5. Scanning Techniques

The assessment uses:

Host Discovery Scan

Purpose:

- Identify available hosts in the network.

Port Scan

Purpose:

- Identify accessible network ports.

Service Detection

Purpose:

- Determine services running on discovered ports.

Version Detection

Purpose:

- Identify software versions for vulnerability analysis.
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6. Risk Evaluation Method

Identified findings are evaluated based on:

Likelihood

Measures the probability that a vulnerability could be exploited.

Rating:

- Low
- Medium
- High

Impact

Measures the potential business effect.

Rating:

- Low
- Medium
- High

Final risk rating:

- Low
 - Medium
 - High
 - Critical
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7. Ethical and Safety Considerations

The assessment follows ethical security testing principles:

- Testing is performed only on authorized systems.
 - No exploitation activities are conducted.
 - No data modification is performed.
 - Results are documented for security improvement purposes.
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8. Expected Output

The scan produces:

- Network scan results
 - Identified services
 - Vulnerability observations
 - Risk ratings
 - Security recommendations
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9. Conclusion

The scan methodology provides a structured approach for evaluating network security posture. The collected information supports vulnerability identification, risk assessment, and development of security recommendations.